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Formal Methods for Cyber-Physical Systems

ASSIGNMENT 1

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Part 1

Assignment

The first part of the assignement request the implementation of the symbolic algorithm for invariant verification. If the test property is not an invariant of the model the algorithm also provide an execution of the system that violate the invariant.

In particular the algorithm must respect the following specification:

- Check if `spec` is an invariant for the loaded model
- The return value is a `tuple` of 2 elements:
  - The first element is a Boolean value that is `True` if the formula is an invariant, `False` otherwise
  - The second element is `None` is the formula is an invariant and contains an execution that violate that specification otherwise.
- The execution take the form of a `tuple` of alternating states and input, ending with a state.

Logic

In order to check if a specification is an invariant of the loaded model we need to check if there exists a path that lead to a state that violate the specification. If such path exists then the specification is not an invariant and we can provide that path as counter-example.

Implementation

The first step is to convert the `spec` into a `bdd` using the provided function `spec_to_bdd(model, spec)`. After that we can use the symbolic algorithm for invariant verification to chek the property.

The basic idea is simple:

1. The algorithm start from the initial states of the model
2. It exmplore the reachable states step by step computing the `post` image of the already reached states. Is important to remove from the `post` image all the states that already have been reached, in order to avoid infinite loops.
3. If one of the newly reached states violate the specification then the property is not an invariant and we can return `False` along with a counter-example.
4. If there are no more new states to explore and none of them violate the specification then the property is an invariant and we can return `True` along with `None`.

```
1  def check_explain_inv_spec(spec):
2      # 1. Setup the bdds
3      model = pynusmv.glob.prop_database().master.bddFsm  # Model fsm
4
5      spec_bdd = spec_to_bdd(model, spec)                 # Bdd of spec
6      negated_spec_bdd = -spec_bdd                        # Bdd of ~spec
7
8      # 2. Explore the regions
9      reach = model.init                                  # Already explored states
10     new = model.init                                    # States to explore
11
```

```
12      trace = [reach]                                    # Trace used to save the execution
13
14      while new.isnot_false() and not new.intersected(negated_spec_bdd):
15
16          new = model.post(new) - reach                  # Remove already checked states
17          reach = reach + new                            # Add new states to reach
18
19          trace.append(new)                              # Add reached region to trace
20
21      if (not new.intersected(negated_spec_bdd)):
22          # The formula is satisfied on every reachable state
23          return (True, None)
24
25      # 3. Build the counter-example
26      invalid_states = new * negated_spec_bdd            # States in new that violate the spec
27      target = model.pick_one_state(invalid_states)      # State that violate the spec
28
29      counter_example = compute_counter_example(model, target, trace)
30
31      # The property is not an invariant, return a counter-example
32      return (False, tuple(counter_example))
```

Counter example generation

If the property is not an invariant we need to provide a counter-example, that is an execution of the system that lead to a state that violate the specification. To do that the algorith saves all the visited states in a list called `trace` and when a violating state is found it uses that list to reconstruct the path from the initial state to the violating state.

Starting from the violating state `s`, the algorithm iterate backwards through the `trace` list, at each step picking one state that can reach the current state in one step. If the model has inputs, the corresponding input is also picked. The process continues until an initial state is reached.

Lookin at the real implementation the function `compute_counter_example(model, s, trace)` does exactly that:

- It first checks if the model has inputs
- It compute the list of invalid states intersecting `new` (the set of states that contains an invalid one) with `neg_spec_bdd`.
- It picks one of those invalid states as starting point and insert it in the counter-example list.
- It then iterate backwards through the `trace` list starting from the second to last element, at each step picking one state that can reach the current state in one step.
- If the model has inputs, the corresponding input is also picked.
- The process continues until the trace is finished.

It's easy to see that the resulting `counter_example` list is a valid execution of the system that lead to a state that violate the specification. Even if the trace is composed by a single region (it can only happen if the system starts with and invalid configuration) the algorithm will work correctly, returning a counter-example composed by the initial state only.

```
1  def compute_counter_example(model, target, trace):
2      current = target                  # Current execution state
3      pre_states = model.pre(current)   # Compute states that can reach current in one step
4      counter_example = [current.get_str_values()]
5
6      # Check if the model uses inputs
7      has_inputs = not not len(model.bddEnc.inputsVars)
8
9      # Interate over the trace starting from the second last
10     for current_set in reversed(trace[:-1]):
11
12         # Get a state that has a transition to current
13         prev_state = model.pick_one_state(current_set * pre_states)
14
15         # Get the input that caused the transition
16         if has_inputs:
17             trans_inputs = model.get_inputs_between_states(prev_state, current)
18             input_values = model.pick_one_inputs(trans_inputs).get_str_values()
19             counter_example.insert(0, input_values)
20         else:
21             counter_example.insert(0, {})
22
23         # Add the picked state
24         counter_example.insert(0, prev_state.get_str_values())
25
26         # Update invariants
27         current = prev_state
28         pre_states = model.pre(current)
29
30     return counter_example
```

Correctness

In this section we will demonstrate in a formal way the correctness of the implemented algorithm. The algoritm is composed by two main loop:

- The first loop explore all the reachable states of the system, checking if one of them violate the specification.

- The second loop generate a counter-example if one of the reachable states violate the specification.

Using the Loop Invariant technique we can prove the correctness of the algorithm.

Exploration of the reachable states

Starting from the first loop, we can indentify three loop invariants:
- `reach`: representing the region of all the already reached states without repetitions.
- `new`: represent the region of all the newly found reachable states.
- `trace`: a list of regions representing all the regions reached at each iteration.

Initially `reach`,`new` and `trace` are initialized with the initial states of the model, therefore the loop invariants hold.

The loop condition check if `new` is not empty, in that case there are still new states to explore and the loop body is executed.

For each interation i:
- `new` is updated to contain all the states that can be reached from `new` in one step, removing all the states that are already in `reach`. Therefore `new` contains only states reachable from the $i-1$ iteration region that are not already in `reach`.
- `reach` is updated to contain all the states reached so far, therefore it contains all the states reachable from the initial states in a finite number of steps.
- `trace` is updated appending the current `new` region, therefore is a list of $i+1$ elements representing all the regions reachable from the previus region.

We can then see that the loop invariants hold for the base case and for the inductive case. Therefore by induction the loop invariants hold for each iteration of the loop.

Since we are dealing with a finite state model the algorithm will eventually explore all the reachable states. Therefore there are only two possible execution:

1. `new` becomes empty, all the reachable states have been explored and none of them violate the specification, so the algorithm return `True` along with `None`.
2. `new` intersected with `neg_spec_bdd` is not empty, there exists at least one newly reached state that violate the specification, so the algorithm return `False` along with a counter-example.

Counter-example generation

In the second part of the algorithm we can identify two loop invariant:

- `counter_example`: a list of alternating states and inputs representing the execution of the system that lead to a state that violate the specification.
- `current_bad_state`: the current state being added to the counter-example list.
- `prev_states`: the set of states that can reach `current_bad_state` in one step.

Initially `current_bad_state` is initialized with a state picked from `new` intersected with `neg_spec_bdd`, therefore it is a state that violate the specification. The `counter_example` list is initialized with that state only. Therefore the loop invariants hold.

The loop iterate backwards through the `trace` list, starting from the second to last element. At each step i:
- `prev_state` is a state picked form the set of states that can reach `current_bad_state` in one step and that are also in the i `trace` region.

- `counter_example` is updated inserting `prev_state` and the corresponding input (if the model has inputs) at the beginning of the list.
- `current_bad_state` is updated to `prev_state`.

The invariants hold for the base case and for the inductive case. Therefore by induction the loop invariants hold for each iteration of the loop.

The trace list is finite, since the model is finite and the algorithm explore only reachable states. Therefore the loop will terminate after a finite number of steps. There are only one possible execution: the trace is finished and the algorithm return the `counter_example` list.

By construction `counter_example` is an execution of the system that lead to a state that violate the specification.

Part 2

Assignment

The second part of the assignment request the implementation of a symbolic algoritm for the verification of a specific class of LTL formulas called *response formulas*. They have a specific form:

$$\square(f \rightarrow \diamondsuit g)$$

Both f and g are Boolean combinations of base formulas with no temporal operators.

In the following pages will be described an implementation of the function `check_examplain_response_spec(spec)` that respect the following specification:

- Check if the given `spec` is satisfied by the loaded SMV model
- The return value is a `tuple` of 2 elemens:
  - The first element is a Boolean value that is `True` if the formula is satisfied, `False` otherwise
  - The second element is `None` is the formula is satisfied but contains a counter-example otherwise.
- The counter-example take the form of a tuple of alternating states and input, ending with a state.
- If the execution is looping the last state should be present somewhere else in the sequence.

Logic

We are dealing with a *response formula* that we can explain such as:

Every time that f happens, g must eventually happen at the same time or afterwards.

Being a reactive property, and therefore happens over an infinite computation, to prove that the formula is not satisfied we must provide a counter-example in witch the negation is true. A path in witch f holds but g never happens again. For that we need 2 parts:

1. **Prefix**: A sequence of states leading to f
2. **Loop**: An infinite cycle in witch g is always `false`.

Implementation

The check of the formula structure was already done by the function `parse(spec)`, so we can just use it and check if the result is a valid `tuple` of f and g or `None`. In the second case the program shuld return `None`, beacause the formula is not a *response formula*.

After that, using the provided function `spec_to_bdd(model, spec)`, we can obtain the `bdd` generated from the two needed formulas: f and $\neg g$.

The algorithm will operate only on the reachable states, so we get them using `model.reachable_states`. From this we can compute the variable `recur` that will contain all the possible starting point for a cycle. In our case all the reachable states in which f holds and $\neg g$ holds.

There are two possibile execution:

1. `recur` becomes empty, there are no possible cycle, the formula holds and the algorithm return a positive result.
2. Otherwise each iteration compute the `pre_reach`, all the state that can be reach `recur` in a finite number of steps greater than 0, and `new`, all the states that might be part of the cycle.

If a cycle is found then the algorithm will build a counter-example using the function `compute_counter_example(model, recur, pre_reach)`, which will be describe in the next section.

The loop runs until there are no more possibile candidates for a cycle.

```
def check_explain_response_spec(spec):
    """
    Return whether the loaded SMV model satisfies or not the GR(1) formula
    `spec`, that is, whether all executions of the model satisfies `spec`
    or not.
    """

    model = pynusmv.glob.prop_database().master.bddFsm
    react = parse(spec)
    if (react == None):
        return None

    f_spec, g_spec = react

    f_bdd = spec_to_bdd(model, f_spec)
    neg_g_bdd = spec_to_bdd(model, ~g_spec)

    reach = model.reachable_states
    recur = reach * neg_g_bdd * f_bdd

    while recur.isnot_false():
        pre_reach = pynusmv.dd.BDD.false()
        new = model.pre(recur) * neg_g_bdd

        while new.isnot_false():
            pre_reach = pre_reach + new
            if recur <= pre_reach:
                return (False, compute_counter_example(model, recur, pre_reach))
            new = (model.pre(new) - pre_reach) * neg_g_bdd
        recur = recur * pre_reach
    return (True, None)
```

Counter example generation

As said before a counter-example is an execution of the system that leads to a state for wich the formula is not satisfied. It is composed of 2 parts:ù

1. **Prefix**: A sequence of states leading to f
2. **Loop**: An infinite cycle in witch g is always `false`.

Those two parts are respectively computed by `compute_prefix` and `compute_loop`.

Build the path

The first computation is easy, we just need to compute a path that reach the starting node of the cycle. To do that, we start from the `model.init` set of states and we proced with the execution until we reach a `frontier` that contains that specific state.

After that we can construct the needed `tuple` visiting the array from the end to the beginning.

```
1  def compute_prefix(model, s):
2    has_inputs = len(model.bddEnc.inputsVars) > 0
3    # 2.3 Reaching the self loop
4
5    # Insert inside frontiers the regions to reach s
6    # [ init ... pre_s ]
7    reached = model.init
8    frontiers = [reached]
9
10   while (reached * s).is_false():
11       next = model.post(reached) - reached
12       frontiers.append(next)
13       reached = reached + next
14
15   path = []
16   last = s
17   # Construct the path traversing the frontiers in reverse
18   for frontier in reversed(frontiers[:-1]):
19       can_reach_last = model.pre(last)
20
21       # The current state is one of the states that can reach last in
22       # one step
23       current_state = model.pick_one_state(frontier * can_reach_last)
24
25       if has_inputs:
26           inputs = model.get_inputs_between_states(current_state, last)
27           path.insert(0, model.pick_one_inputs(inputs).get_str_values())
28       else:
29           path.insert(0, {})
30
31       path.insert(0, current_state.get_str_values())
32
33       last = current_state
34
```

```
35    return path
```

Build the loop

The loop is more complex. We need to compute a loop starting from `s`, following the path through the frontiers list.

We first need to identify the smallest number of steps `k` that contains `s`. We then define 2 invariant:

- `path` -> All the possible path ending in `s` in `k` steps.
- `current` -> Last state put in the `path`.

The algoritm build the counter-example iterating backwards the `frontiers` taking one state and its corresponding input at the time.

After the last iteration `path` contains a full path ending in `s` that starts from the first `frontier` region.

Unite those 2 path togheter leads to a path starting from an initial state, reaching a state in witch f holds and $\neg g$ holds, and lead to a cycle in witch $\neg g$ holds.

We can then be sure that the counter-example is built on the right way and rapresent a valid execution of the system that don't satisfies the formula.

```
1  def compute_loop(model, s, frontiers):
2    has_inputs = len(model.bddEnc.inputsVars) > 0
3
4    # Phase 2.2: Building the self loop
5    for k in range(len(frontiers)):
6        if s <= frontiers[k]:
7            break
8    # frontiers[k] contains the loop knot
9
10   path = [ s.get_str_values() ]
11   curr = s
12
13   for i in range(k - 1, -1, -1):
14
15       old = curr
16       pred = model.pre(curr) * frontiers[i]
17       curr = model.pick_one_state(pred)
18
19       if has_inputs:
20           inputs = model.get_inputs_between_states(old, curr)
21           path.insert(0, model.pick_one_inputs(inputs).get_str_values())
22       else:
```

```
            path.insert(0, {})

        path.insert(0, curr.get_str_values())

    # Looping input
    if has_inputs:
        inputs = model.get_inputs_between_states(s, curr)
        path.insert(0, model.pick_one_inputs(inputs).get_str_values())
    else:
        path.insert(0, {})

    path.insert(0, s.get_str_values())
    return path
```